

Development of a Functional Annotation and Taxonomic Classification Workflow for Metagenome-assembled genomes

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Abstract. Shotgun metagenomics has significantly advanced microbial ecology by enabling the reconstruction of Metagenome-Assembled Genomes (MAGs) directly from complex environmental samples. However, the taxonomic and functional characterization of these multi-kingdom communities faces important methodological challenges. Although a wide array of bioinformatic tools and some integrated workflows are currently available, choosing the optimal combination of tools for a comprehensive analysis is not straightforward. Furthermore, many established pipelines are still heavily biased towards prokaryotes. Consequently, the eukaryotic fraction (e.g., fungi and protists) is frequently neglected or misclassified due to its higher genomic complexity and the scarcity of standardized multi-kingdom integration solutions. To address this gap, this project aims to develop a modular, scalable, and reproducible bioinformatic pipeline dedicated to the integrated taxonomic classification and functional annotation of multi-kingdom MAGs. Initially, a theoretical evaluation of the underlying methodologies of state-of-the-art domain-specific tools (e.g., GTDB-Tk, Prokka, MetaEuk) will be performed. Subsequently, a Snakemake-based architecture will be developed to orchestrate the annotation of pre-classified contigs, seamlessly directing them through tailored multi-kingdom modules. Designed to seamlessly integrate into a larger metagenomics pipeline, this workflow will generate standardized functional and taxonomic profiles. These formatted outputs will directly support downstream community metabolic modeling, advancing our understanding of complex microbiomes.

Keywords: Shotgun Metagenomics, Metagenome-Assembled Genomes, Multi-kingdom Microbiome, Bioinformatic Workflows, Functional Annotation

1 Contextualization and Motivation

Metagenomics has fundamentally transformed microbiome research by enabling the cultivation-independent analysis of microbial communities directly from environmental or host-associated samples [25]. The transition from targeted amplicon sequencing (e.g., 16S rRNA) to high-throughput shotgun metagenomics has been particularly revolutionary [13]. By sequencing the entire genetic material

within a sample, shotgun metagenomics provides unparalleled insights into the taxonomic composition and functional potential of complex ecosystems in all domains of life, including bacteria, archaea, viruses, and eukaryotes. Crucially, this approach facilitates the computational reconstruction of Metagenome-Assembled Genomes (MAGs), allowing researchers to resolve the microbial identity at the strain level and directly link specific taxa to their metabolic capabilities [13].

However, in natural and engineered environments, microorganisms do not exist in isolation, prokaryotes and eukaryotes cohabit and continuously interact, shaping the overall ecosystem function [20,25]. Despite the inherent multi-kingdom nature of these communities, a significant methodological limitation persists: comprehensive workflows that also analyze eukaryotic organisms remain notably scarce [27].

While various tools are available for extracting biological information from eukaryotic genomes, determining the optimal combination of software for accurate taxonomic classification and functional annotation remains a significant challenge. Eukaryotic MAGs are typically larger and possess highly complex genomic architectures characterized by the presence of introns, alternative splicing mechanisms, extensive non-coding regions, and high densities of repetitive elements [3,24]. Consequently, applying standard, prokaryote-optimized annotation pipelines to whole-community datasets frequently results in the misclassification of eukaryotic sequences, the generation of false-positive gene predictions, or the complete exclusion of the eukaryotic fraction. Therefore, there is a critical need to develop integrative and modular analytical frameworks capable of segregating and independently processing multi-kingdom MAGs [6,12,24], ensuring that downstream analyses such as community level metabolic modeling are built upon accurate and comprehensive data [27].

2 Aims

The primary objective of this project is to design, implement, and validate a modular and scalable bioinformatics workflow dedicated to integrated taxonomic classification and functional annotation of genomes from distinct biological kingdoms. This pipeline aims to standardize the processing of both prokaryotic and eukaryotic organisms within a single automated framework, effectively overcoming current fragmentation in metagenomic data analysis. To achieve this overarching goal, the project is structured around three interconnected objectives.

First, the research will conduct a comprehensive evaluation of computational tools in the current literature for domain segregation, quality control, and functional annotation. Following this methodological review, the second objective focuses on developing and testing a highly reproducible architecture capable of automatically routing preseggregated contigs into tailored annotation modules specialized for their respective biological domains. Ultimately, the final objective is to ensure that the outputs generated provide deep insights into microbiome composition while meeting the strict quality requirements necessary to serve as

direct inputs for downstream systems biology applications, specifically allowing the subsequent reconstruction of Genome-scale Metabolic Models (GEMs).

3 State-of-the-art

3.1 The Concept of Metagenomics and MAGs

This section outlines the foundational concepts of modern microbiome analysis. It begins by defining the principles of metagenomics, subsequently exploring how advanced sequencing technologies enable the computational reconstruction of individual genomes directly from complex environmental samples.

Metagenomics is defined as the culture-independent analysis of genetic material recovered directly from environmental or host-associated samples. By bypassing the need for isolation and cultivation in the laboratory, metagenomic approaches enable the study of complex microbial communities in their native contexts. This capability enables simultaneous capture of the taxonomic composition and functional potential of the community as a whole [25]. Consequently, metagenomics has established itself as a central methodology in microbiome research, with wide-ranging applications in environmental sciences, biotechnology, agriculture, and human health.

The true power of this approach has been cemented by the adoption of Next-Generation Sequencing (NGS) technologies, which form the basis of shotgun metagenomics [4,8,18]. By sequencing the entirety of Deoxyribonucleic acid (DNA) extracted from a sample, shotgun metagenomics enables a comprehensive analysis that encompasses all domains of life (bacteria, archaea, viruses, and eukaryotes). More importantly, this technique enables the computational reconstruction of complete or partial genomes from a mixture of sequences. These reconstructed genomes, known as MAGs, have become the fundamental unit in genome-level genomics, enabling the direct detection of genes, operons, and metabolic pathways of individual populations within the ecosystem [13].

3.2 Overview of Multi-Kingdom Genomic Analysis

This section outlines the computational challenges inherent to analyzing samples containing multiple biological domains. It contrasts the genomic complexities of prokaryotes and eukaryotes, evaluates the limitations of current integration strategies, and establishes the need for modular routing architectures.

Shotgun metagenomics has the major advantage of capturing all DNA in a sample, including bacteria, archaea, and eukaryotes [13]. However, analyzing this mixture of different kingdoms of life in a single process poses a major computational challenge. The central problem lies in the fact that most current bioinformatic tools have been designed and optimized solely for prokaryotic organisms, whose genomes are typically more compact and simpler.

In contrast, the genomes of eukaryotes, such as fungi and protists, are much more complex and challenging. They are substantially larger and contain specific

structures, such as introns, vast non-coding regions, and alternative splicing mechanisms [7]. If we apply the algorithms developed for bacteria directly to these eukaryotic genomes, the results will be incorrect. The system tends to make serious errors, such as interpreting parts of a single gene with introns as if they were several independent genes. This generates a large number of “false positives” and completely distorts conclusions about the metabolic pathways of the community.

While specialized software has been successfully developed to annotate these complex eukaryotic genomes independently, the primary obstacle in modern metagenomics remains the integration of these distinct biological domains. Although specific tools have been developed to perform this integration, they present significant compromises. Current approaches typically fall into two categories. The first employs generalized unified workflows, such as SqueezeMeta [21]. Although these pipelines offer the advantage of streamlined execution by processing all data simultaneously, they frequently suffer from a strong prokaryotic bias, relying on generalized algorithms that fail to capture complex eukaryotic structures. The second category involves modular sequential processing, using platforms like Anvi'o [3] or MetaWRAP [22]. While these platforms allow for high accuracy by attempting to merge specialized tools, they require intense manual labor, demand significant computational resources, and often struggle to harmonize incompatible output formats effectively. Consequently, the study of ecological interactions between different biological kingdoms remains severely complicated.

To address these integration challenges, it is becoming increasingly clear that all DNA within a sample cannot be treated uniformly. Overcoming current obstacles requires a paradigm shift towards modular architectures capable of ingesting genomes that have been previously classified by biological domain. Developing workflows that can act as dedicated routing engines seamlessly directing pre-segregated prokaryotic and eukaryotic sequences to their respective specialized annotation tools represents a critical gap in the current literature. Filling this gap is essential to maximize the biological accuracy of comprehensive multi-kingdom microbiome studies.

3.3 Taxonomic and Functional Annotation Strategies

This section details the distinct analytical strategies required for the taxonomic classification and functional annotation of separated genomic data. It first examines the highly optimized workflows utilized for prokaryotic organisms, subsequently exploring the sophisticated computational tools necessary to resolve complex eukaryotic structures. Finally, the section evaluates current comprehensive pipelines attempting to integrate these domains, highlighting their respective advantages and limitations.

Following the successful segregation of MAGs by biological domain [24], the next step involves a thorough characterization of their genetic content. In the context of metagenomics, this process unfolds along two fundamental axes: taxonomic classification, which identifies evolutionary lineages, and functional an-

notation, which elucidates metabolic potential. Given the marked divergence in genomic architecture between prokaryotes and eukaryotes, it is imperative to adopt independent and specialized analytical strategies for each group.

Prokaryotic Genomes Regarding the prokaryotic domains of bacteria and archaea, bioinformatic workflows are currently highly optimized. While taxonomic classification is frequently anchored by GTDB-Tk [2] as the established methodological standard, alternative frameworks, including the Bin Annotation Tool [23] and PhyloPhlAn [1] provide robust complementary strategies for phylogenetic profiling. Instead of relying on a single marker gene such as 16S rRNA, these modern approaches use the alignment of dozens of universal marker proteins, enabling the precise placement of the MAGs within a robust phylogenetic tree topologically consistent with the Genome Taxonomy Database (GTDB). Concerning functional annotation, although Prokka [16] has historically stood out for its efficiency, contemporary pipelines such as Bakta [15] and DRAM [17] are increasingly adopted. By taking advantage of the compact genomic architecture of bacteria, specifically the general absence of introns, these tools integrate fast-running *ab initio* gene prediction algorithms [5] with extensively curated databases. Notably, Bakta offers enhanced feature resolution, whereas DRAM is specifically designed to profile metabolic potential, generating comprehensive functional profiles that directly support downstream systems biology applications [16].

Eukaryotic Genomes In contrast, the annotation of eukaryotic MAGs, namely fungi and microeukaryotes, requires a substantially more sophisticated computational approach. Traditional predictive tools often struggle with the discontinuity of eukaryotic genes fragmented by introns. To overcome this limitation, modern workflows rely on a combination of specialized solutions. For structural and functional annotation, MetaEuk [7] offers a dynamic homology-based strategy to accurately identify coding exons and infer intron boundaries. Alternatively, comprehensive workflows like EukMetaSanity [11] utilize *ab initio* gene predictors, including Augustus [19], to resolve these complex genomic architectures. Beyond gene prediction, ensuring the overall validity of eukaryotic MAGs demands rigorous quality control and lineage assessment. Estimating genome completeness and contamination is a crucial validation step effectively managed by dedicated tools such as EukCC [14] and BUSCO [9].

Ultimately, while the independent annotation of eukaryotic and prokaryotic fractions is well established, the primary obstacle lies in their seamless integration. Currently, researchers rely on a few comprehensive pipelines that attempt to process communities with multiple domains, but these solutions present distinct compromises between computational efficiency and biological accuracy. For instance, unified pipelines such as SqueezeMeta [21] offer rapid joint assembly profiling, but they typically rely on generalized gene prediction strategies (e.g.,

Prodigal) that may not fully capture the complex structural features of eukaryotic genomes [7]. Other approaches utilize modular sequential processing, such as MetaWRAP [22], which provides an excellent framework for prokaryotic binning and refinement, yet its native support for eukaryotes remains limited due to its reliance on prokaryote-specific tools like CheckM and Prokka [14]. In contrast, highly modular platforms like Anvi'o [3] provide robust and dedicated workflows for both domains, although comparative reviews indicate that their interactive nature often demands significant computational resources and extensive manual orchestration [26].

Table 1. Critical comparison of current bioinformatics pipelines for the integration of multiple biological domains.

Pipeline	Integration Strategy	Key Advantages	Major Limitations
SqueezeMeta [21]	Unified joint assembly and homology search.	Highly automated; fast execution; simultaneously profiles prokaryotes and eukaryotes.	Shallow structural annotation for eukaryotes (e.g., relies on prokaryotic gene callers); relies heavily on reference databases [7].
MetaWRAP [22]	Modular sequential processing.	Excellent for prokaryotic binning and refinement; highly flexible framework.	Eukaryotic support is limited (reliance on CheckM and Prokka) and often requires unofficial custom adaptations [14].
Anvi'o [3]	Highly modular and interactive architecture.	Deep analysis for both domains; excellent visualization capabilities.	Steep learning curve; demands intensive computational resources and manual orchestration [26].

4 Preliminary Methodology

Based on the challenges identified in the current literature regarding the integration of multiple biological kingdoms and to directly address the established objectives of this project, we propose a modular, scalable, and highly reproducible bioinformatic pipeline. The architecture and data flow of this proposed methodology are visually represented in Figure 1. The workflow begins with ingestion of the initial MAGs in FASTA format. To ensure that each sequence is analyzed with the most biologically appropriate tools, the first critical phase is accurate domain segregation. The pipeline utilizes Tiara [6], a tool that employs advanced algorithms based on sequence composition to classify input data. This process allows the system to accurately sort and branch the contigs into distinct prokaryotic and eukaryotic pathways before any structural annotation begins.

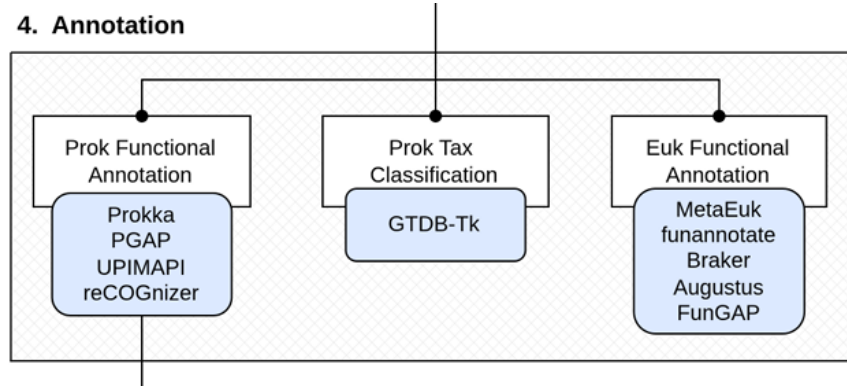


Fig. 1. Overview of workflow

To orchestrate this complex parallel processing, the system adopts Snakemake [10] as its core workflow management system. Written in Python, Snakemake defines the workflow by precisely mapping the dependency relationships between input files and generated files through a directed acyclic graph. This modular architecture ensures automatic task parallelization. To guarantee full reproducibility of the analysis across diverse computational infrastructures and to prevent software dependency conflicts, Snakemake is natively integrated with containerization technologies like Docker and package managers like Conda. In addition, the pipeline leverages wrappers maintained by the bioinformatics community. Using these rigorously tested components, the workflow minimizes custom scripting, reduces human error potential, and solidifies reliability through highly standardized execution protocols.

Following the initial segregation, the contigs undergo parallel processing within their respective domain modules. Each branch incorporates dedicated quality control and annotation tools optimized for specific genomic architectures. For the prokaryotic branch, the pipeline integrates standard industry tools such as GTDB-Tk [2] for precise taxonomic classification and Prokka for rapid functional annotation. On the other hand, the eukaryotic branch utilizes BUSCO to evaluate genome completeness and MetaEuk to accurately predict complex gene structures and assign functional profiles, effectively overcoming the discontinuity of eukaryotic genes. Ultimately, the final stage of the pipeline consolidates the processed data from both parallel branches. The end result is the generation of standardized taxonomic and metabolic profiles formatted into unified data tables. These consolidated outputs are perfectly structured for direct integration into downstream computational applications, specifically enabling the reconstruction and simulation of GEMs for entire biological communities.

References

1. Asnicar, F., Thomas, A.M., Beghini, F., Mengoni, C., Manara, S., Manghi, P., Zhu, Q., Bolzan, M., Cumbo, F., May, U., et al.: Precise phylogenetic analysis of microbial isolates and genomes from metagenomes using phylophlan 3.0. *Nature communications* **11**(1), 2500 (2020)
2. Chaumeil, P.A., Mussig, A.J., Hugenholtz, P., Parks, D.H.: Gtdb-tk: a toolkit to classify genomes with the genome taxonomy database (2020)
3. Eren, A.M., Esen, Ö.C., Quince, C., Vineis, J.H., Morrison, H.G., Sogin, M.L., Delmont, T.O.: Anvi'o: an advanced analysis and visualization platform for 'omics data. *PeerJ* **3**, e1319 (2015)
4. Hu, T., Chitnis, N., Monos, D., Dinh, A.: Next-generation sequencing technologies: An overview. *Human immunology* **82**(11), 801–811 (2021)
5. Hyatt, D., Chen, G.L., LoCascio, P.F., Land, M.L., Larimer, F.W., Hauser, L.J.: Prodigal: prokaryotic gene recognition and translation initiation site identification. *BMC bioinformatics* **11**(1), 119 (2010)
6. Karlicki, M., Antonowicz, S., Karnkowska, A.: Tiara: deep learning-based classification system for eukaryotic sequences. *Bioinformatics* **38**(2), 344–350 (2022)
7. Levy Karin, E., Mirdita, M., Söding, J.: Metaeuk—sensitive, high-throughput gene discovery, and annotation for large-scale eukaryotic metagenomics. *Microbiome* **8**(1), 48 (2020)
8. Liu, L., Li, Y., Li, S., Hu, N., He, Y., Pong, R., Lin, D., Lu, L., Law, M.: Comparison of next-generation sequencing systems. *BioMed research international* **2012**(1), 251364 (2012)
9. Manni, M., Berkeley, M.R., Seppey, M., Simão, F.A., Zdobnov, E.M.: Busco update: novel and streamlined workflows along with broader and deeper phylogenetic coverage for scoring of eukaryotic, prokaryotic, and viral genomes. *Molecular biology and evolution* **38**(10), 4647–4654 (2021)
10. Mölder, F., Jablonski, K.P., Letcher, B., Hall, M.B., van Dyken, P.C., Tomkins-Tinch, C.H., Sochat, V., Forster, J., Vieira, F.G., Meesters, C., et al.: Sustainable data analysis with snakemake. *F1000Research* **10**, 33 (2025)
11. Neely, C.J., Hu, S.K., Alexander, H., Tully, B.J.: The high-throughput gene prediction of more than 1,700 eukaryote genomes using the software package euk-metasanity. *bioRxiv* pp. 2021–07 (2021)
12. Pronk, L.J., Medema, M.H.: Whokaryote: distinguishing eukaryotic and prokaryotic contigs in metagenomes based on gene structure. *Microbial genomics* **8**(5), 000823 (2022)
13. Ranjan, R., Rani, A., Metwally, A., McGee, H.S., Perkins, D.L.: Analysis of the microbiome: Advantages of whole genome shotgun versus 16s amplicon sequencing. *Biochemical and biophysical research communications* **469**(4), 967–977 (2016)
14. Saary, P., Mitchell, A.L., Finn, R.D.: Estimating the quality of eukaryotic genomes recovered from metagenomic analysis with eukcc. *Genome biology* **21**(1), 244 (2020)
15. Schwengers, O., Jelonek, L., Dieckmann, M.A., Beyvers, S., Blom, J., Goesmann, A.: Bakta: rapid and standardized annotation of bacterial genomes via alignment-free sequence identification. *Microbial genomics* **7**(11), 000685 (2021)
16. Seemann, T.: Prokka: rapid prokaryotic genome annotation. *Bioinformatics* **30**(14), 2068–2069 (2014)
17. Shaffer, M., Borton, M.A., McGivern, B.B., Zayed, A.A., La Rosa, S.L., Solden, L.M., Liu, P., Narrowe, A.B., Rodríguez-Ramos, J., Bolduc, B., et al.: Dram for

- distilling microbial metabolism to automate the curation of microbiome function. *Nucleic acids research* **48**(16), 8883–8900 (2020)
18. Slatko, B.E., Gardner, A.F., Ausubel, F.M.: Overview of next-generation sequencing technologies. *Current protocols in molecular biology* **122**(1), e59 (2018)
 19. Stanke, M., Keller, O., Gunduz, I., Hayes, A., Waack, S., Morgenstern, B.: Augustus: ab initio prediction of alternative transcripts. *Nucleic acids research* **34**(suppl_2), W435–W439 (2006)
 20. Sun, Z., Li, G., Wang, C., Jing, Y., Zhu, Y., Zhang, S., Liu, Y.: Community dynamics of prokaryotic and eukaryotic microbes in an estuary reservoir. *Scientific reports* **4**(1), 6966 (2014)
 21. Tamames, J., Puente-Sánchez, F.: Squeezemeta, a highly portable, fully automatic metagenomic analysis pipeline. *Frontiers in microbiology* **9**, 3349 (2019)
 22. Uritskiy, G.V., DiRuggiero, J., Taylor, J.: Metawrap—a flexible pipeline for genome-resolved metagenomic data analysis. *Microbiome* **6**(1), 158 (2018)
 23. Von Meijenfeldt, F.B., Arkhipova, K., Cambuy, D.D., Coutinho, F.H., Dutilh, B.E.: Robust taxonomic classification of uncharted microbial sequences and bins with cat and bat. *Genome biology* **20**(1), 217 (2019)
 24. West, P.T., Probst, A.J., Grigoriev, I.V., Thomas, B.C., Banfield, J.F.: Genome-reconstruction for eukaryotes from complex natural microbial communities. *Genome research* **28**(4), 569–580 (2018)
 25. Wooley, J.C., Godzik, A., Friedberg, I.: A primer on metagenomics. *PLoS computational biology* **6**(2), e1000667 (2010)
 26. Yepes-García, J., Falquet, L.: 2pipe starts with a question: matching you with the correct pipeline for mag reconstruction. *mSystems* pp. e00844–25 (2026)
 27. Zorrilla, F., Buric, F., Patil, K.R., Zelezniak, A.: metagem: reconstruction of genome scale metabolic models directly from metagenomes. *Nucleic acids research* **49**(21), e126–e126 (2021)